

Experiment 12

Objective: Switch faults study in User Interface section: Vibrator/ Buzzer/ Speaker/ Hands-free

Apparatus required:

- 5G VoLTE Smart Phone Training Platform
- SIM card(s) of any 5G service provider supporting 5G frequency band
- Power Supply for Training Platform Scientech 2140 with Mains Cord
- Hands free kit

Operating Mode:

5G VoLTE Smart Phone Training system works in two operating mode.

- i) 'Adaptor Mode'
- ii) 'Battery Mode'

Mode can be selected by a 'Mode Selection Switch' provided on the 'Training Platform' at top left side, below DC Input socket and Power button.

- **Adaptor Mode (Switch Pressed):** In this mode the 'Training Platform' operates on external Power Supply.
- **Battery Mode (Switch Released):** In this mode the 'Training Platform' operates with Battery supplied (if charging level is >90%). Also the battery can be charged in this mode.

Operating Conditions:

Before performing this experiment, please ensure the following;

- Insert SIM card(s) before switching 'On' the 'Training Platform' **(Most important point to be noted)**
- Also ensure that the SIM cards in use should be of any 5G service provider supporting 5G frequency band.
- All the Switch Faults should be in 'On' position (fault-free condition).
- To operate in 'Battery' mode, Battery must be charged more than 90%.

Procedure:

The following experiments can be performed in both operating mode; i.e. in 'Adaptor' mode or in 'Battery' mode.

- a) To operate in 'Adaptor' mode, refer the procedure given in **Experiment 1**.
- b) To operate in 'Battery' mode, refer the procedure given in **Experiment 3**.

Note: It is recommended to operate these experiments in 'Adaptor' mode to reduce the Battery usage.

- Once the set up is ready to operate as per the procedure given in the recommended **Experiment 11**, perform the experiment as follows.
 - Check that the SIM is detected with sufficient network strength.
 - Now the platform is ready for further experiments as follows.
1. Insert the switch fault as per the table given below. Observe the effect and find the description of the fault given. Measure/observe the voltages/signals at respective 'Test Points' of 'User Interface' section with respect to the nearest 'Ground' test point.

Note: Insert only one fault at a time and after observation 'Reset' it to its original position before inserting the next switch fault.

Switch Number	Fault Number	Effect/ Action	Description
SW4	1	No Effect	Not connected in circuit.
SW4	2	MIC of Hands-free stops to function.	Signals to Microphone's 'MIC+' pin disconnected.
SW4	3	'Speaker - R' of Hands-free stops to function	Signals to Speaker's 'Right' pin disconnected.
SW4	4	MIC and Speaker (L/R) all stops to function	Common Ground of Hands-free section disconnected.
SW4	5	Speaker (L/R) Stops to function	Signals to Speaker's 'Left/Right' pin disconnected.
SW4	6	'Speaker - L' of Hands-free stops to function	Signals to Speaker's 'Left' pin disconnected.
SW4	7	'Vibrator' stops to function.	Vibrator driver voltage 'VIB+' (2.8V) disconnected from circuit. Reset fault to resume.
SW4	8	'Vibrator' stops to function.	Ground signal 'VIB-' (0V) disconnected from vibrator. Reset fault to resume.
SW4	9	Speaker (Earpiece) stops to function.	Audio signals disconnected from Speaker. Reset fault to resume.
SW4	10	'Buzzer' stops to function.	PWM signals disconnected from Buzzer. Reset fault to resume.

Conclusion: Switch Fault insertion in 'User Interface' section and its effects along with the possible cause are studied and functioning of the respective signal/voltage is understood.

Experiment 13

Objective: Observing constellations of I/Q signals of Trans-receiver section

Apparatus required:

- 5G VoLTE Smart Phone Training Platform
- SIM card(s) of any 5G service provider supporting 5G frequency band
- Power Supply for Training Platform Scientech 2140 with Mains Cord
- Oscilloscope

Operating Mode:

5G VoLTE Smart Phone Training system works in two operating mode.

- i) 'Adaptor Mode'
- ii) 'Battery Mode'

Mode can be selected by a 'Mode Selection Switch' provided on the 'Training Platform' at top left side, below DC Input socket and Power button.

- **Adaptor Mode (Switch Pressed):** In this mode the 'Training Platform' operates on external Power Supply.
- **Battery Mode (Switch Released):** In this mode the 'Training Platform' operates with Battery supplied (if charging level is >90%). Also the battery can be charged in this mode.

Operating Conditions:

Before performing this experiment, please ensure the following;

- Insert SIM card(s) before switching 'On' the 'Training Platform' **(Most important point to be noted)**
- Also ensure that the SIM cards in use should be of any 5G service provider supporting 5G frequency band.
- All the Switch Faults should be in 'On' position (fault-free condition).
- To operate in 'Battery' mode, Battery must be charged more than 90%.

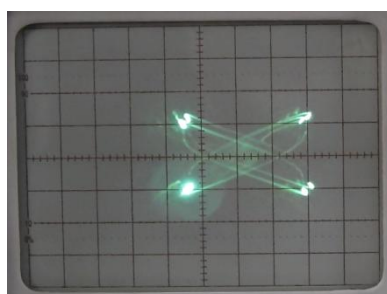
Procedure:

The following experiments can be performed in both operating mode; i.e. in 'Adaptor' mode or in 'Battery' mode.

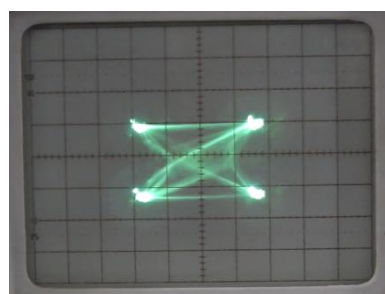
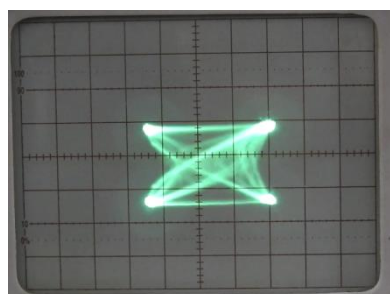
- a) To operate in 'Adaptor' mode, refer the procedure given in **Experiment 1**.
- b) To operate in 'Battery' mode, refer the procedure given in **Experiment 3**.

Note: It is recommended to operate these experiments in 'Adaptor' mode to reduce the Battery usage.

1. Once the set up is ready to operate as per the procedure given in the recommended **Experiment 11**, perform the experiment as follows.
2. Check the charging status and the % level of Battery at the Display of Smart Phone (If operating in 'Battery' mode, charging should be more than 90%).
3. Check that the SIM is detected with sufficient network strength.
4. Set both the trace (CH1 and CH2) of Oscilloscope at the center.
5. Switch the Oscilloscope in 'XY Mode' to observe the signal constellations of I/Q signals.
6. Connect the two probes to Oscilloscope's CH1 and CH2 inputs and connect their grounds.
7. Make a call from any other phone to the 'Training Platform'.
8. When the dialed number rings, do not attend the incoming call.
9. Connect CH1 probe at test points I(Rx) and CH2 probe to Q(Rx) of 'RF Trans-receiver' section.
10. Adjust both CH1 and CH2 attenuators to get the constellation of RX I/Q signals as per below image.



11. Similarly, connect CH1 probe at test points I(Tx) and CH2 probe to Q(Tx) of 'RF Trans-receiver' section.
12. Adjust both CH1 and CH2 attenuators to get the constellation of TX I/Q signals as per below image.



Conclusion: Constellations of TX I/Q signals and RX I/Q signals of 'RF Trans-receiver' section is observed in XY mode of Oscilloscope.

Experiment 14

Objective: Observing the spectrum of Trans-receiver RF carrier signal

Apparatus required:

- 5G VoLTE Smart Phone Training Platform
- SIM card(s) of any 5G service provider supporting 5G frequency band
- Power Supply for Training Platform Scientech 2140 with Mains Cord
- PC/Laptop

Operating Mode:

5G VoLTE Smart Phone Training system works in two operating mode.

- i) 'Adaptor Mode'
- ii) 'Battery Mode'

Mode can be selected by a 'Mode Selection Switch' provided on the 'Training Platform' at top left side, below DC Input socket and Power button.

- **Adaptor Mode (Switch Pressed):** In this mode the 'Training Platform' operates on external Power Supply.
- **Battery Mode (Switch Released):** In this mode the 'Training Platform' operates with Battery supplied (if charging level is >90%). Also the battery can be charged in this mode.

Operating Conditions:

Before performing this experiment, please ensure the following;

- Insert SIM card(s) before switching 'On' the 'Training Platform' **(Most important point to be noted)**
- Also ensure that the SIM cards in use should be of any 5G service provider supporting 5G frequency band.
- All the Switch Faults should be in 'On' position (fault-free condition).
- To operate in 'Battery' mode, Battery must be charged more than 90%.

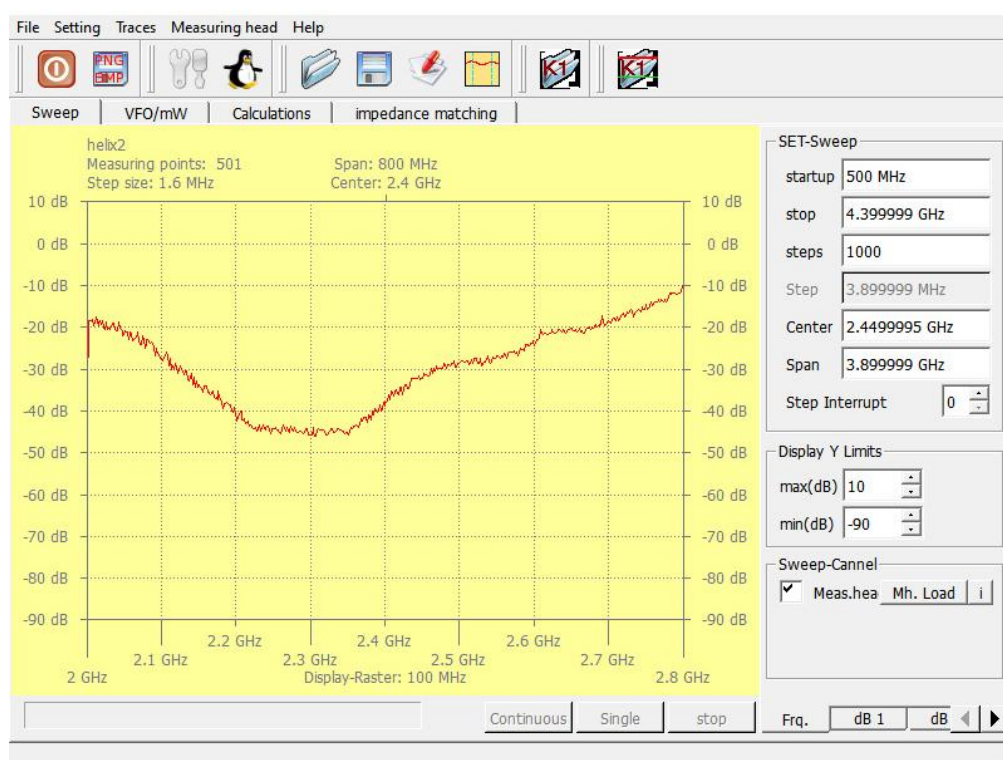
Procedure:

The following experiments can be performed in both operating mode; i.e. in 'Adaptor' mode or in 'Battery' mode.

- a) To operate in 'Adaptor' mode, refer the procedure given in **Experiment 1**.
- b) To operate in 'Battery' mode, refer the procedure given in **Experiment 3**.

Note: It is recommended to operate these experiments in 'Adaptor' mode to reduce the Battery usage.

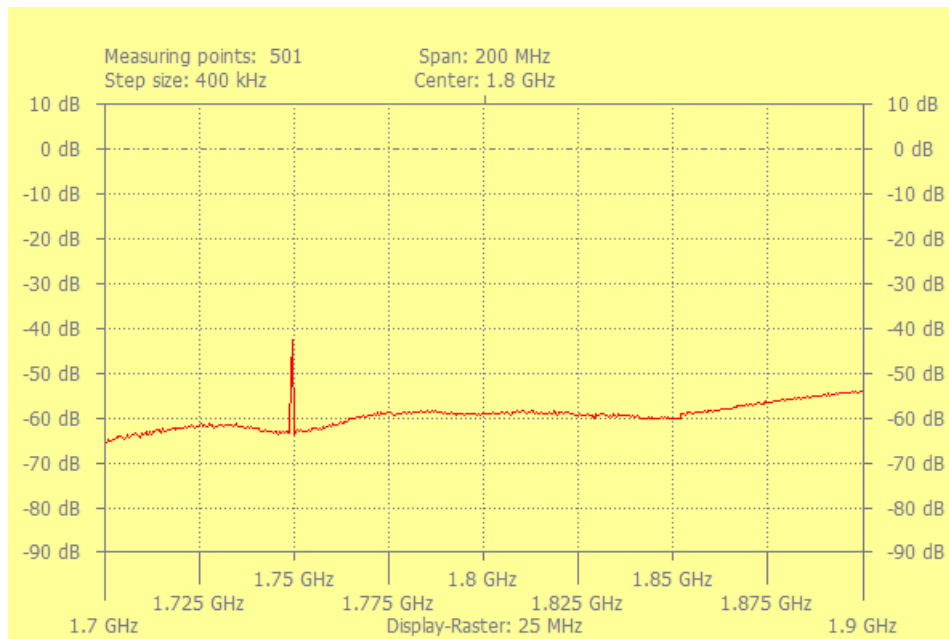
1. Once the set up is ready to operate as per the procedure given in the recommended **Experiment 11**, perform the experiment as follows.
2. Check the charging status and the % level of Battery at the Display of Smart Phone (If operating in 'Battery' mode, charging should be more than 90%).
3. Check that the SIM is detected with sufficient network strength.
4. Install the software for PC based 'RF Spectrum Analyzer' by running the .exe file supplied with product.
5. Connect USB cable between 'RF Interface' Input port of 'RF Spectrum Analyzer' and USB port of PC.
6. Now open the software. 'On' LED will start to blink. Wait till the port detection message appears.
7. Following window will appear on the screen of the PC monitor.



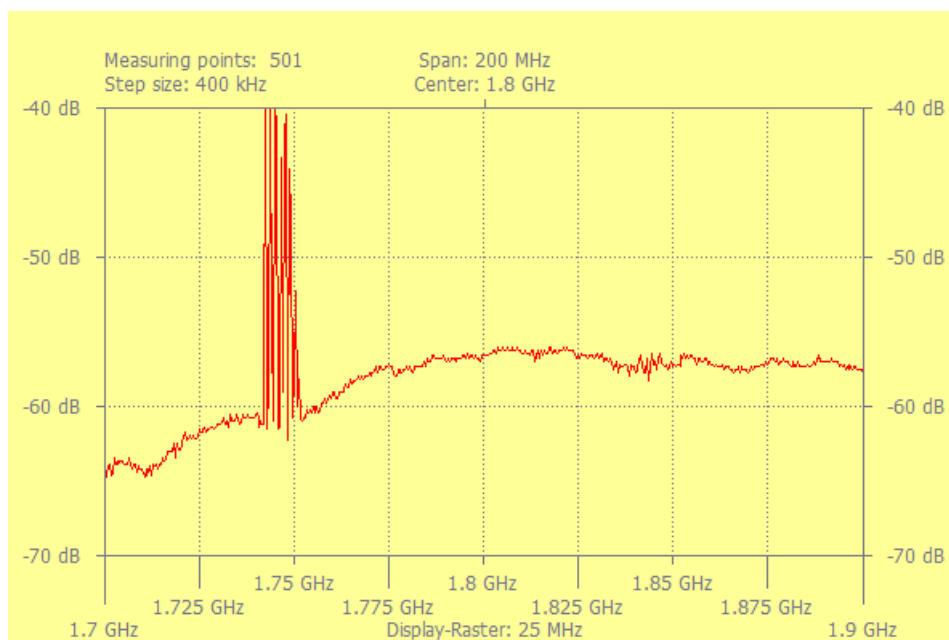
8. Press 'Reset' button to refresh the window.
9. Set the 'Startup' and 'Stop' frequencies and steps accordingly. The Center frequency and Span will be set automatically. Set the Display Y Limits max(dB) and min(dB). And Click Start button.
10. Make a call from any other phone to the 'Training Platform'.

Example: If the received frequency is expected to be around 1.8 GHz, we can select Startup frequency 1.7 GHz and Stop frequency 1.9 GHz and Steps can be

set to 500 or more for better capture of frequency. And initially the Display Y Limits can be set max(dB) to 0dB and min(dB) to -90dB. Below response will be there.

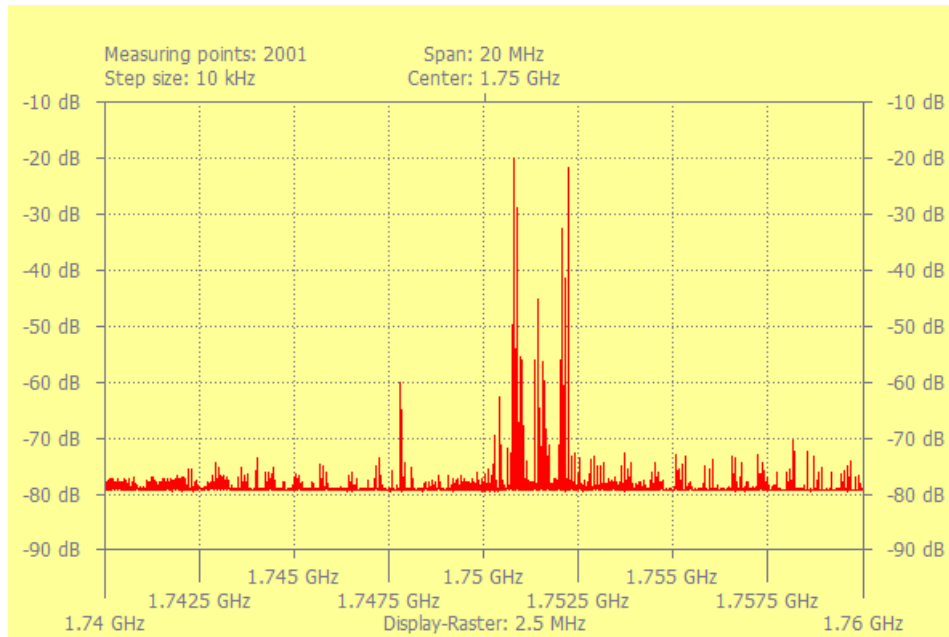


11. Now to have elaborated spike of amplitude, click stop button and adjust the Display Y Limits to max(dB) to -40dB and min(dB) to -70dB and click start to capture the signal again.



12. Now to have a better resolution of spectrum of the received frequency, click stop and adjust the frequency span to a narrow band i.e. the startup and stop

frequency to 1.74 GHz to 1.76 GHz respectively. And set Display Y Limits to max(dB) to -20dB and min(dB) to -80dB and click start to capture the signal.



13. The study of signal can be done for various combinations of Frequency span, Startup and Stop frequency, Steps and Display Y Limits.
14. The Spectrum file can be saved from the File menu of the software.
15. For detailed features of the software menu, refer the User manual of RF Spectrum Analyzer supplied with this product.